Hyperspectral Image Recovery Using Nonconvex Sparsity and Low-Rank Regularizations

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***Abstract*— Hyperspectral image (HSI) restoration is an important preprocessing step in HSI data analysis to improve the image quality for subsequent applications of HSI. In this article, we introduce a spatial–spectral patch-based nonconvex sparsity and low-rank regularization method for HSI restoration. In contrast to traditional approaches based on convex penalties or nonconvex spectral penalty alone, we consider the sparsity of HSI in the spatial–spectral domain and combine the nonconvex low-rank penalty and the nonconvex 3-D total variation (TV)-like sparsity regularization to fully exploit the correlations in both spatial–spectral dimensions of the HSI data set. In addition, we propose a fast iterative variable splitting-based algorithm to effectively solve the corresponding optimization problem. Numerical experiments on both simulated and real HSI data sets demonstrate that the proposed nonconvex low-rank and TV (NonLRTV) method significantly improves the recovered image quality compared with the state-of-the-art algorithms.**

***Index Terms*— Hyperspectral image (HSI), nonconvex low-rank and TV (NonLRTV), restoration.**

I. Introduction

HYPERSPECTRAL imaging deals with the collection of images acquired simultaneously from across the electromagnetic spectrum, which consists a few hundreds of narrow spectral band images. Since each ground cover substance can be represented by a specific spectral curve in hyperspectral imagery, hyperspectral image (HSI) has found a wide range of applications in remote sensing such as identification and detection of the ground surface, mineral exploration, agriculture, geosciences, astronomy, military surveillance [1], [2]. However, due to the physical limitations of the hyperspectral imaging sensors, the interference during the transmission of the image data, and the photon effects, HSIs collected in practice are often unavoidably suffered from different types of degradations, such as Gaussian noise, stripes, impulse noise, and missing data [3]. Thus, HSI recovery is a significant preprocessing step in practical applications in order to obtain more useful images for further applications.

So far, there are different methods introduced for HSI data recovery. A straightforward denoising method is to exploit the spatial information or the spectral information by considering the HSI data set as 2-D or 1-D signals, respectively. For example, the singular value decomposition (SVD) [4], block-matching 3-D filtering [5], and the nonlocal based method [6] are adopted to restore the HSI band-by-band. However, these methods ignore the correlations between neighboring pixels or different spectral bands, which are not able to provide optimal results. Recently, algorithms that exploit both the spatial and spectral information are introduced to improve the image quality. Chen and Qian [7] proposed an HSI denoising method by combining the principal component analysis (PCA) transform and the wavelet shrinkage. In [8], the HSI data sets are recovered by fusing the denoising results in the spectral domain and the spatial domain. A hybrid spatial–spectral wavelet shrinkage denoising scheme was proposed in [9], which benefits from the dissimilarity of the signal regularity in both the spatial and the spectral domain of the HSI data set.

In recent years, low-rank regularized optimization models have been widely applied in image processing and computer vision [10], [11]. For HSI data sets, the spectral profiles of the neighboring voxels are highly correlated, and the spectrally adjacent bands represent strong correlations, which reveal the low-rank structure of the HSI. Inspired by this fact, several low-rank models have recently been adopted in HSI restoration. PCA is a widely applied scheme which utilizes the low-rank prior [12], where the HSI data set is mapped into a linear combination of a series of uncorrelated principal components by the orthogonal transformation. However, PCA is sensitive to outliers and non-Gaussian noise, such as deadlines and stripes. In order to solve the problem, Wright *et al.* [13] and Candès *et al.* [11] proposed robust PCA (RPCA), where the data can be decomposed into a low-rank matrix and a sparse outlier matrix, and thus can be recovered with a high probability under certain incoherence conditions. Based on this model, a low-rank matrix recovery (LRMR) method was developed for HSI restoration [3], where the HSI is divided into patches and the LRMR is applied on each patch to remove the mixed noise simultaneously. Similarly, a noise-adjusted iterative low-rank matrix approximation method (NAILRMA) [14] was introduced considering different noise intensities in different

Manuscript received December 7, 2018; revised March 29, 2019 and July 22, 2019; accepted August 15, 2019. Date of publication September 17, 2019; date of current version December 27, 2019. This work was supported in part by the National Natural Science Foundation of China under Grant 61871159, in part by the Natural Science Foundation of Heilongjiang Province under Grant F2016018, in part by the Natural Scientific Research Innovation Foundation in Harbin Institute of Technology, and in part by the National Natural Science Foundation of Key International Cooperation under Grant 61720106002. *(Corresponding author: Yanfeng Gu.)*

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Digital Object Identifier 10.1109/TGRS.2019.2937901

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bands. In addition, algorithms based on the low-rank tensor recovery [15], [16] and low-rank factorization have also been proposed [17]. However, these methods only focus on exploiting the spectral correlation of the HSI data set. In addition, the matrix rank has to be approximated as prior for LRMR and NAILRMA methods, which is not easy to acquire. In [18], a joint spatial and spectral low-rank (SSLR) regularization method was proposed, which simultaneously accounts for the low-rank property across the spectral domain and the nonlocal low-rank property over the spatial domain.

Since Rudin *et al.* [19] proposed the total variation (TV) model, TV-based methods have been widely applied in image processing due to its capability of removing noise and preserving sharp edges [20], [21]. Recently, TV regularizations have also been well applied to HSI recovery. A spectral–spatial adaptive hyperspectral TV (SSAHTV) denoising algorithm was proposed [22] where both the spectral and the spatial differences were considered. A similar spatial–spectral TV method was also proposed in [23]. Chang *et al.* [24] introduced a simultaneous destriping and denoising method by using the unidirectional TV and sparse representations. In [25], a cubic TV (CTV) method was proposed incorporating the 2-D TV for spatial domain and 1-D TV for the spectral domain.

In order to exploit the spectral low-rank property and the spatial piecewise smooth structure of the HSI data set, He *et al.* [26] developed a TV-regularized low-rank matrix factorization method low-rank and TV (LRTV) integrating the nuclear norm and the spatial TV constraints. However, LRTV uses the band-by-band TV regularization, while ignoring the spectral consistency and smoothness in the spectral domain. In addition, similar to LRMR method, the LRTV scheme requires the rank to be estimated as prior, which is difficult to achieve in practice. The TV regularization was integrated with the weighted nuclear norm minimization [27], [28] by Wu *et al.* [29]. Chen *et al.* [30] proposed using the tensor nuclear norm and the TV regularization to exploit the low-rank prior and the spatial–spectral continuity of color and multispectral images. In [31], the tensor Tucker decomposition method is combined with an anisotropic spatial–spectral TV constraint to capture the spectral correlation and the piecewise smooth structure in spatial–spectral domain. In [32], 3-D TV regularization was combined with the low-rank constraint to effectively restore the HSI data set.

All of the aforementioned models are based on convex regularizations. Although the convex constraint method has been successfully applied in HSI restoration, the l_1 penalty yields inconsistent results under some cases [33], [34]. In addition, the l_1 regularization is not capable of restoring the optimal original signal with the least measurements [35]. Recently, the nonconvex schemes based on l_p penalties have attracted considerable attention and have been successfully applied in many imaging application fields [36], [37]. Xie *et al.* [38] proposed an iterative HSI restoration method for mixture noise removal based on LRMA model [weighted Schatten p-norm (WSN)-LRMA], where the weighted nonconvex Schatten-p norm rank constraint was used. Chen *et al.* [39] also introduced a nonconvex low-rank matrix approximation method, referred to as NonLRMA, which was based on a novel nonconvex r-norm regularizer to remove the mixture noise in HSI data sets and obtain satisfactory results.

In this article, we propose a combined nonconvex regularization method, where both the nonconvex spectral and sparsity penalties are used to effectively remove the mixed noise in HSI. Specifically, we adapt a 3-D nonconvex TV regularization to further characterize the piecewise smoothness in both spatial and spectral domains. By using the nonconvex sparsity penalty, the large image values are less penalized, which leads to better preservation of edges and image details. Besides, the nonconvex Schatten p-norm is applied, which is capable of providing a more sparse approximation of the singular values and thus better exploiting the low-rank property of the HSI data set. In order to solve the corresponding optimization problem, we develop an effective fast iterative algorithm based on the half-quadratic splitting method [40] for TV regularization [41], [42]. In order to accelerate the convergence, we adapt the continuation scheme, which prevents the algorithm from trapping into local optimum.

In summary, the contributions of this article are as follows.

1) A nonconvex low-rank regularization model is proposed to restore the clean HSIs from the degraded observation contaminated by mixed types of noise. The proposed model rearranges the 3-D HSI data patch as 2-D matrix, so that the spatial and spectral correlation between different HSI bands can be better exploited.
2) A designed nonconvex spatial–spectral TV-like sparsity regularization is incorporated into the low-rank regularized framework. Unlike the traditional spatial–spectral TV penalty, the proposed nonconvex regularization is capable of obtaining a more sparse solution, which enables more accurate restoration of the features and details of the HSIs. Compared with the traditional convex regularization, the proposed NonLRTV method is able to provide better restoration results theoretically and practically.
3) In order to solve the local minimum problem, which is common when solving the nonconvex optimization model, we propose a fast variable splitting-based iterative algorithm. Specifically, we rewrite both nonconvex regularizations with quadratic functions using variable splitting, which leads to a constrained minimization problem. The constrained problem can thus be solved using an alternating minimization method with an analytic solution.

The rest of this article is organized as follows. We briefly introduce the related background literature to make this article self-contained in Section II. In Section III, we propose the NonLRTV model and the corresponding optimization scheme for HSI restoration. We investigate the performance of the proposed algorithm in the experimental results for simulated data and real data in Section IV, and the conclusions are drawn in Section V.

II. Background

The degradation of the HSI data set can be described as follows [22]:

$$\mathbf{y} = \mathbf{h} + \mathbf{n} \tag{1}$$

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where $\mathbf{y}$ is the observed degraded HSI, $\mathbf{h}$ is the clean image that requires restoration, and $\mathbf{n}$ represents the mixed noise including the stripes, deadlines, and the Gaussian noise. All the data sets are 3-D cubes with the size of $M \times N \times K$ with $M \times N$ the spatial dimension and K the number of spectral bands.

In order to preserve the local features, we decompose the HSI data set into overlapped spatial–spectral 3-D patches with the size of $m \times m \times K$. For a 3-D patch centered at (i, j), the degradation model can be expressed as follows:

$$y(i, j) = h(i, j) + n(i, j) \tag{2}$$

where $y(i, j)$, $h(i, j)$, and $n(i, j)$ are the patches corresponding to $\mathbf{y}$, $\mathbf{h}$, and $\mathbf{n}$, respectively.

The spatial–spectral 3-D patch $h(i, j)$ can be rearranged as a Casorati matrix with the size of $m^2 \times K$

$$\mathbf{H} = \begin{bmatrix} h(\mathbf{x}_0, s_0) & \cdots & h(\mathbf{x}_0, s_{K-1}) \\ \vdots & & \\ h(\mathbf{x}_{m^2-1}, s_0) & \cdots & h(\mathbf{x}_{m^2-1}, s_{K-1}) \end{bmatrix} \tag{3}$$

where the ith column represents the vectorized observed image at the spectral band i, and the jth row corresponds to the spectral variations of the pixel $\mathbf{x}_j$. Thus, the degradation process can be modeled in the following matrix form:

$$\mathbf{Y} = \mathbf{H} + \mathbf{N} \tag{4}$$

where $\mathbf{Y}$ and $\mathbf{N}$ are the Casorati matrices obtained from $y(i, j)$, $n(i, j)$, which are with the same size of $\mathbf{H}$. Since the spectral information of the neighboring pixels are highly correlated, the matrix $\mathbf{H}$ presents the low-rank property, namely, the rank of $\mathbf{H}$ is given by $r < \min(m^2, K)$.

In order to exploit the spectral low-rank structure of the clean HSI data set $\mathbf{Y}$, we can restore $\mathbf{Y}$ by solving the following constrained optimization problem:

$$\hat{\mathbf{H}} = \arg\min_{\mathbf{H}} \|\mathbf{H} - \mathbf{Y}\|^2 \quad \text{s.t. } \operatorname{rank}(\mathbf{H}) \le r. \tag{5}$$

The above optimization problem can be rewritten using Lagrange multipliers as

$$\hat{\mathbf{H}} = \arg\min_{\mathbf{H}} \|\mathbf{H} - \mathbf{Y}\|^2 + \lambda \operatorname{rank}(\mathbf{H}). \tag{6}$$

Naturally, using the l_0-norm penalty, i.e., $\|\cdot\|_0$, as the low-rank relaxation, gives the exact formulation to minimize the above regularization problem [11], [13]. However, the l_0-norm makes the problem NP-hard and nonconvex, thus intractable to solve. A common approach is to use the convex relaxation, namely, the nuclear norm, to replace the nonconvex rank penalty. With the relaxation, the restoration of $\mathbf{H}$ becomes

$$\hat{\mathbf{H}} = \arg\min_{\mathbf{H}} \|\mathbf{H} - \mathbf{Y}\|^2 + \lambda \|\mathbf{H}\|_* \tag{7}$$

where $\|\mathbf{H}\|_* = \sum_i \sigma_i$ represents the nuclear norm of $\mathbf{H}$, which is the sum of the singular values of $\mathbf{H}$, given by $\mathbf{H} = \mathbf{U}\Sigma\mathbf{V}^*$; and λ is the balancing parameter of the two penalty terms.

III. Proposed Nonconvex Low-Rank and Total Variation Method for HSI Recovery

A. Problem Formulation

In order to fully explore the structure in spatial and spectral domains of the HSI, we combine the nonconvex low-rank penalty and the nonconvex TV penalty to obtain a novel NonLRTV model to restore the HSI data set. Thus, both the spectral correlation and the gradient sparsity property in the spatial–spectral domain of HSI data set can be effectively captured. The proposed NonLRTV model is formulated as

$$\min \lambda_1 \mathcal{J}_1(\mathbf{H}) + \lambda_2 \mathcal{J}_2(\mathbf{H}) \quad \text{s.t. } \|\mathbf{H} - \mathbf{Y}\|^2 \le \epsilon \tag{8}$$

where $\mathcal{J}_1(\mathbf{H})$ indicates the low-rank term, $\mathcal{J}_2(\mathbf{H})$ represents the sparsity term, and λ_1 and λ_2 are the balancing parameters. Equation (8) can be rewritten using Lagrange multipliers as follows:

$$\hat{\mathbf{H}} = \arg\min_{\mathbf{H}} \|\mathbf{H} - \mathbf{Y}\|^2 + \lambda_1 \mathcal{J}_1(\mathbf{H}) + \lambda_2 \mathcal{J}_2(\mathbf{H}). \tag{9}$$

In the above optimization problem, the relaxation of the low-rank constraint $\mathcal{J}_1(\mathbf{H})$ is chosen as the nonconvex Schatten-p norm penalties, specified as

$$\mathcal{J}_1(\mathbf{H}) = \|\mathbf{H}\|_{l_{p_1}}^{p_1} = \sum_{i=1}^{\min(m^2, K)} \sigma_i^{p_1}. \tag{10}$$

Here, $\mathbf{H} = \mathbf{U}\Sigma\mathbf{V}^*$ is the SVD of $\mathbf{H}$, and $\Sigma = \operatorname{diag}([\sigma_1, \sigma_1, \ldots, \sigma_r])$. Note that if $p_1 = 1$, the spectral penalty becomes the nuclear norm. When $p < 1$, this penalty becomes nonconvex.

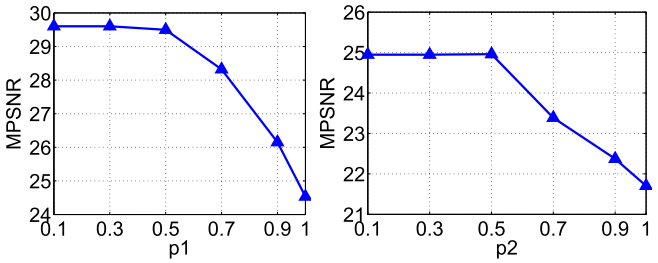
It has been proven that l_1 norms often introduce extra bias in estimation [43]. Moreover, the l_1 minimization is not able to obtain results that are sparse enough, which makes the recovered images suboptimal. Compared with the l_1 norm, the nonconvex l_p norm gives more accurate approximation of the l_0 norm. A phase diagram study has been conducted in [44], which demonstrates that l_p norms can generate more sparse solutions than l_1 norms. In addition, as the value p decreases, l_p norms yield sparser solutions. As a result, the nonconvex seminorms are able to considerably improve the restoration performance of the signal, compared with traditional l_1 norms [45], [46].

In order to exploit the piecewise smoothness along the spectral–spatial dimension, we propose to use the gradient sparsity regularization, which is specified as the nonconvex TV-like penalty

$$\mathcal{J}_2(\mathbf{H}) = \left\| \sqrt{\sum_{i=0}^{q-1} |\mathcal{S}_i(\mathbf{H})|^2} \right\|_{l_{p_2}}^{p_2} \tag{11}$$

where $\mathcal{S}_i(\mathbf{H}) = \mathbf{R}_i^H \mathbf{H} \mathbf{C}_i$, $\mathbf{R}_i$ and $\mathbf{C}_i$ are operators that transform the row-space and column-space of $\mathbf{H}$ into the sparsity domain, respectively. The nonconvex TV penalty is thus given by the above expression when $q = 3$, $\mathbf{R}_0 = \mathbf{D}_x$, $\mathbf{C}_0 = \mathbf{I}$, $\mathbf{R}_1 = \mathbf{D}_y$, $\mathbf{C}_1 = \mathbf{I}$, $\mathbf{R}_2 = \mathbf{I}$, $\mathbf{C}_2 = \mathbf{D}_s$, where $\mathbf{D}_x$, $\mathbf{D}_y$, $\mathbf{D}_s$ are the finite difference operators along the x, y, and the spectral dimension s. If $p_2 = 1$, (11) becomes the standard convex 3-D TV regularization. In this article, we choose $p_2 < 1$ to

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(a) (b)

Fig. 1. Illustration of the advantages of using nonconvex penalties. (a) MPSNR values of the restored 3-D patch using nonconvex and convex low-rank penalties. (b) MPSNR values of the restored 3-D patch using nonconvex and convex TV penalties.

obtain a nonconvex regularization. Thus, the edges and the fine details in HSI can be better preserved.

In Fig. 1, we illustrate the advantages of using nonconvex penalties by separately applying the low-rank penalty and the 3-D TV penalty, with different p values, to a 3-D patch cropped from the HSI data set used in the simulated data experiment in Section IV-A1. We plot the evolution of the mean peak signal-to-noise ratio (MPSNR) values of the restored HSI patch with respect to p values using only the low-rank penalty and using only the 3-D TV penalty in Fig. 1(a) and (b), respectively. It is observed that the convex penalties ($p_1 = 1$ and $p_2 = 1$) yield the lowest MPSNR values. Nevertheless, when the nonconvex penalties ($p_1 < 1$ and $p_2 < 1$) are applied, the results get improved significantly. We observe that when $p \in (0.5, 1)$, as the p value decreases, we obtain higher MPSNRs; and when $p \in (0, 0.5)$, the results show no significant difference. These findings are consistent with the theory in [44], which further demonstrate that the nonconvex penalties outperform their convex counterparts in recovering images with the inherent property of sparsity.

B. Optimization Algorithm

To realize an efficient algorithm to solve (9), we introduce a fast iterative majorize–minimize algorithm based on the half-quadratic minimization scheme. In particular, we rewrite the optimization function as a constrained minimization problem using variable splitting method

$$\begin{aligned} &\arg\min_{\mathbf{H}} \ \|\mathbf{H}-\mathbf{Y}\|^2 + \lambda_1 \mathcal{J}_1(\mathbf{P}) + \lambda_2 \left\| \sqrt{\sum_{i=0}^{2} |\mathbf{T}_i|^2} \right\|_{l_{p_2}}^{p_2} \\ &\text{s.t. } \mathbf{H} = \mathbf{P} \ \ \mathbf{T}_i = \mathcal{S}_i(\mathbf{H}), \quad i = 0, 1, 2 \end{aligned} \tag{12}$$

where $\mathbf{P}$ and $\mathbf{T}_i, i = 0, 1, 2$ are the auxiliary variables. Thus, the regularization penalties can be majorized using quadratic functions, specified as

$$\mathcal{J}_1(\mathbf{H}) = \min_{\mathbf{P}} \mathcal{J}_1(\mathbf{P}) + \frac{\beta_1}{2}\|\mathbf{H}-\mathbf{P}\|^2 \tag{13}$$

$$\mathcal{J}_2(\mathbf{H}) = \min_{\mathbf{T}_i, i=0,1,2} \left\| \sqrt{\sum_{i=0}^{2} |\mathbf{T}_i|^2} \right\|_{l_{p_2}}^{p_2} + \frac{\beta_2}{2} \sum_{i=0}^{2} \|\mathcal{S}_i(\mathbf{H}) - \mathbf{T}_i\|^2. \tag{14}$$

Using the majorization functions (13) and (14), the original optimization function can be rewritten as follows:

$$\{\mathbf{H}, \mathbf{P}, \mathbf{T}_i\} = \arg\min_{\{\mathbf{H},\mathbf{P},\mathbf{T}_i\}} \mathcal{J}_{\beta_1,\beta_2}(\mathbf{H}, \mathbf{P}, \mathbf{T}_i) \tag{15}$$

where

$$\begin{aligned} &\mathcal{J}_{\beta_1,\beta_2}(\mathbf{H}, \mathbf{P}, \mathbf{T}_i) \\ &= \|\mathbf{H}-\mathbf{Y}\|^2 + \lambda_1 \mathcal{J}_1(\mathbf{P}) + \frac{\lambda_1\beta_1}{2}\|\mathbf{H}-\mathbf{P}\|^2 \\ &\quad + \lambda_2 \left\| \sqrt{\sum_{i=0}^{2} |\mathbf{T}_i|^2} \right\|_{l_{p_2}}^{p_2} + \frac{\lambda_2\beta_2}{2} \sum_{i=0}^{2} \|\mathcal{S}_i(\mathbf{H}) - \mathbf{T}_i\|^2. \end{aligned} \tag{16}$$

Here, when $\beta_1, \beta_2 \to \infty$, (16) tends to the original optimization function (9). However, when we choose β_1 and β_2 to be large values, the convergence rate is low. In order to overcome this tradeoff, we propose to use the continuation scheme [47], which will be illustrated in detail later.

We adopt an alternating minimization scheme to solve the above optimization problem. In particular, we alternatively minimize each of the three variables $\mathbf{H}$, $\mathbf{P}$, and $\mathbf{T}_i$, assuming the other two variables to be known. We denote the nth iteration of the variables as $\mathbf{H}_n$, $\mathbf{P}_n$, and $\mathbf{T}_{i,n}$, respectively.

C. $\mathbf{H}$ Subproblem: Minimization With Respect to $\mathbf{H}$, Assuming $\mathbf{P}$ and $\mathbf{T}_i$ to be Known

We first derive $\mathbf{H}_{n+1}$, assuming $\mathbf{P} = \mathbf{P}_n$,$\mathbf{T}_i = \mathbf{T}_{i,n}$. Thus, the cost function becomes

$$\begin{aligned} \mathbf{H}_{n+1} = \arg\min_{\mathbf{H}} &\|\mathbf{H}-\mathbf{Y}\|^2 + \frac{\lambda_1\beta_1}{2}\|\mathbf{H}-\mathbf{P}_n\|^2 \\ &+ \frac{\lambda_2\beta_2}{2} \sum_{i=0}^{2} \left\|\mathcal{S}_i(\mathbf{H}) - \mathbf{T}_{i,n}\right\|^2. \end{aligned} \tag{17}$$

Since (17) is a quadratic function with respect to $\mathbf{H}$, we can obtain its minimizer as

$$\begin{aligned} &\left(2\mathbf{I} + \lambda_1\beta_1\mathbf{I} + \lambda_2\beta_2 \sum_{i=0}^{2} \mathcal{S}_i^T \mathcal{S}_i\right) \mathbf{H}_{n+1} \\ &\qquad = 2\mathbf{Y} + \lambda_1\beta_1\mathbf{P}_n + \lambda_2\beta_2 \sum_{i=0}^{2} \mathcal{S}_i^T \mathbf{T}_{i,n} \end{aligned} \tag{18}$$

where $\mathbf{I}$ is the identity matrix. Since the variables on the right-hand side of (18) are assumed to be known, the solution to the linear system of equations can be obtained. In this article, we efficiently solve the above equation using iterative conjugate gradient algorithm.

D. $\mathbf{P}$ Subproblem: Minimization With Respect to $\mathbf{P}$, Assuming $\mathbf{H}$ and $\mathbf{T}_i$ to be Known

Based on (15) and (16), assuming $\mathbf{H} = \mathbf{H}_n$, the minimization of $\mathbf{P}$ can be expressed as

$$\mathbf{P}_{n+1} = \arg\min_{\mathbf{P}} \frac{\beta_1}{2}\|\mathbf{H}_n - \mathbf{P}\|^2 + \mathcal{J}_1(\mathbf{P}). \tag{19}$$

Here, if the low-rank penalty $\mathcal{J}_1(\mathbf{P})$ is the convex nuclear norm, namely, $p_1 = 1$, the solution to (19) can be easily

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obtained following the iterative singular value thresholding algorithm [48], given by

$$\mathbf{P}_{n+1} = \sum_{i=1}^{\min(m^2,K)} \max\left(\sigma_i - \frac{1}{\beta_1}, 0\right) \mathbf{u}_i \mathbf{v}_i^H \tag{20}$$

where $\mathbf{H}_{n+1} = \sum_{i=1}^{\min(m^2,K)} \sigma_i \mathbf{u}_i \mathbf{v}_i^H$ is the SVD of $\mathbf{H}_{n+1}$. In this article, when the nonconvex low-rank penalty is used ($p_1 < 1$), similarly as addressed in [49], the shrinkage step of the Schatten-p norm can be derived as follows:

$$\mathbf{P}_{n+1} = \sum_{i=1}^{\min(m^2,K)} \max\left(\sigma_i - \frac{\sigma_i^{(p_1-1)}}{\beta_1}, 0\right) \mathbf{u}_i \mathbf{v}_i^H. \tag{21}$$

E. $\mathbf{T}_i$ Subproblem: Minimization With Respect to $\mathbf{T}_i$, Assuming $\mathbf{H}$ and $\mathbf{P}$ to be Known

Based on (15) and (16), assuming $\mathbf{H} = \mathbf{H}_n$, the minimization of $\mathbf{T}_i$ is given by

$$\mathbf{T}_{i,n+1} = \arg\min_{\mathbf{T}_i} \frac{\beta_2}{2} \sum_{i=0}^{2} \|\mathcal{S}_i(\mathbf{H}_n) - \mathbf{T}_i\|^2 + \left\| \sqrt{\sum_{i=0}^{2} |\mathbf{T}_i|^2} \right\|_{l_{p_2}}^{p_2}. \tag{22}$$

In this subproblem, it involves solving all of the components for $\mathcal{S}_i(\mathbf{H})$ in order to apply the shrinkage algorithm to the magnitude of the nonconvex TV term, namely, $(\sum_{i=0}^{2} |\mathcal{S}_i(\mathbf{H}_n)|^2)^{1/2}$. Similar problems have been addressed for iterative algorithms for TV minimization [42] and its nonconvex variants [50]. Consequently, we can derive the shrinkage rule for (22) as follows.

1) When the regular convex TV penalty is used ($p_2 = 1$)

$$\mathbf{T}_{i,n+1} = \frac{\mathcal{S}_i(\mathbf{H}_n)}{\mathbf{Q}} \max\left(\mathbf{Q} - \frac{1}{\beta_2}, 0\right) \tag{23}$$

where $\mathbf{Q} = (\sum_{i=0}^{2} |\mathcal{S}_i(\mathbf{H}_n)|^2)^{1/2}$.

2) For the nonconvex TV penalty ($p_2 < 1$)

$$\mathbf{T}_{i,n+1} = \frac{\mathcal{S}_i(\mathbf{H}_n)}{\mathbf{Q}} \max\left(\mathbf{Q} - \frac{\mathbf{Q}^{(p_2-1)}}{\beta_2}, 0\right). \tag{24}$$

The convergence of the aforementioned three-step alternating minimization scheme has been well proved in [51]. However, due to the tradeoff between the convergence rate and the accuracy of the approximation of the original cost function by the majorization function, we propose to use the continuation scheme during the minimization steps. Specifically, we start with small initial values for β_1 and β_2 to make the algorithm converge fast to the optimal solution, while gradually increasing the values. When β_1 and β_2 are with large values, the majorization function approximates the optimization problem (9) well, which guarantees the final solution. The algorithm is terminated till the relative change of the cost function for successive iterations is less than a predefined tolerance value, which is given by

$$r = \left| \frac{\mathcal{J}_n - \mathcal{J}_{n-1}}{\mathcal{J}_n} \right| < \varepsilon. \tag{25}$$

Algorithm 1 NonLRTV Algorithm ($\mathbf{Y}$, λ_1, λ_2)

```
Initialization: N ← 1, β1 ← β_init1, β2 ← β_init2 ;
while N < N_max do
    n ← 1;
    while n < n_max do
        Update H based on (18) using conjugate gradient steps;
        Update P using the shrinkage rule (21);
        Update T_i using the shrinkage rule (24);
        if The convergence rule (25) is met then
            Break;
        else
            n ← n + 1
        end
    end
    β1 ← β1*β_inc1;
    β2 ← β2*β_inc2;
    N ← N + 1
end
Output: H
```

To summarize, the pseudocode for the proposed NonLRTV method is shown in Algorithm 1.

In NonLRTV algorithm, the input variables are the noisy HSI $\mathbf{Y}$, the balancing parameters λ_1 and λ_2, the continuation parameters β_1 and β_2, and the convergence parameter r. The output is the clean HSI data $\mathbf{H}$. There are two loops in the algorithm, where the parameters β_1 and β_2 are increased in the outer iterations, and the cost function (16) is minimized in the inner iterations. In this article, we initialize β_1 and β_2 with 5, and increase them by $\beta_{inc1} = \beta_{inc2} = 5$ in each outer iteration. We observe that the continuation scheme significantly improves the convergence rate while guarantees the accuracy of the final results. The number of the outer iterations N_{max} is chosen to be 10, and the maximum inner iteration n_{max} is 20, the convergence threshold r is set to be 10^{-6}. We typically use around 35 conjugate gradient steps for convergence. The balancing parameters λ_1 and λ_2 are tuned in each case to obtain the optimal result.

IV. Experimental Results and Discussion

In order to demonstrate the utility of the proposed method, we conduct both simulated and real HSI data experiments for HSI restoration. We compare the performance of proposed NonLRTV algorithm with several state-of-the-art approaches, including the LRMR method [3], the TV-regularized low-rank matrix factorization (LRTV) [26], the NAILRMA method [14], the WSN-based LRMA (WSN-LRMA) [38], and the nonconvex NonLRMA model [39]. The implementation codes are provided by the authors. The gray values of each HSI band are scaled to [0,1] before restoration. We perform the experiments in MATLAB R2014a on a computer with a CORE i5 3.3-GHz CPU and 8-GB memory. The parameters of the compared methods for the following experiments are manually optimized according to their default strategies under all scenarios.

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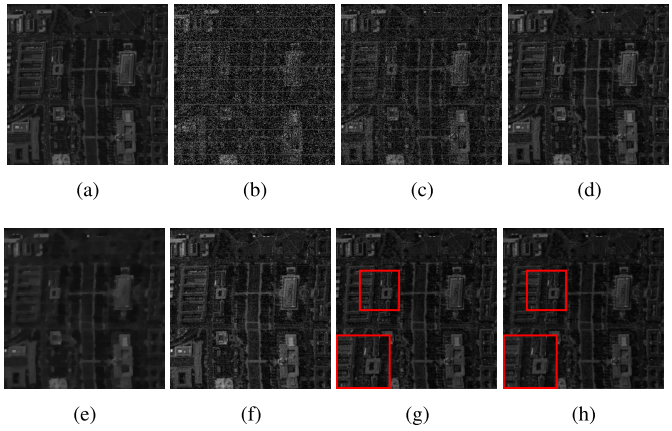


Fig. 2. Restoration results on simulated experiment: Washington DC Mall. (a) Original image. (b) Noisy image. Restored images obtained by (c) LRMR, (d) NAILRMA, (e) LRTV, (f) WSN-LRMA, (g) NonLRMA, and (h) proposed NonLRTV.

A. Simulated Data Experiments

1) Washington DC Mall: The hyperspectral digital imagery collection experiment (HYDICE) image of the Washington DC Mall is used in the first simulated experiment. The size of the data set is $1208 \times 307 \times 191$ [52]. Due to the page limitation, we select a subregion of the data set with the size of $256 \times 256 \times 15$, which is used as the ground truth. In the experiment, we add two types of typical noise in HSI, described in detail as follows.

1) Zero-mean Gaussian noise with different variance values were added to all bands of HSI. The Gaussian noise variance is randomly selected in the range from 0.06 to 0.17. The mean variance value of all the bands is 0.1.
2) Periodical stripes are added to five different bands (bands 2, 5, 7, 10, and 12) with width of 1 pixel. The number of stripes in each band is 12.

The restoration results of band 2 are shown in Fig. 2. We observe that the LRMR and LRTV methods are not able to restore the image effectively. The NAILRMA scheme performs better in removing the stripes, while the Gaussian noise is still obvious in the recovered image. Compared with the convex regularized methods, the nonconvex approaches obtain satisfactory results. However, the Gaussian noise can still be observed in the restored images by the WSN-LRMA and the NonLRMA schemes. The proposed NonLRTV outperforms the other methods in removing both of the Gaussian noise and the stripes while effectively preserving the details of the image. In order to quantitatively evaluate the results, the MPSNR index and the mean structural similarity index (MSSIM) of the restored images are reported in Table I, shown in the first two rows. In addition, we report the peak signal-to-noise ratio (PSNR) and structural similarity index (SSIM) measurements for three different bands. The results indicate that the proposed NonLRTV method consistently performs the best, with around 10-dB improvement over the noisy image.

2) Pavia City Center: The second HSI data set for the simulated data experiment is the Pavia city center data set, which was collected by the Reflective Optics System Imaging Spectrometer (ROSIS-03). After removing the first 22 bands which are heavily contaminated by noise, we use a subregion of the image with the size of $200 \times 200 \times 80$ [53]. In order to investigate the proposed algorithm in removing mixed noise, we add four types of noise in the Pavia city center data set, specified as follows.

1) Zero-mean Gaussian noise with different variance values was added to all bands of HSI. The variance of the Gaussian noise is randomly selected in the range from 0 to 0.1.
2) Impulse noise was added to 11 selected bands, from band 30 to band 40. The percentage of the impulse noise was 0.1.
3) Deadlines were simulated on five selected bands, from band 50 to band 54. The width of the deadlines ranges from 1 to 3 pixels.
4) Stripes were added on four selected bands, from band 61 to band 64. The width of the stripes ranges from 1 to 3 pixels. The number of stripes randomly changes from 3 to 10.

Fig. 3 displays the restoration results of the Pavia data set for band 38, which was contaminated by the Gaussian noise and the impulse noise. We observe that the restored image by the LRMR method has obvious noise remained. The LRTV method is capable of removing most of the noise, whereas the image details are blurred in the restored image. NAILRMA, WSN-LRMA, and NonLRMA methods perform better with the image details preserved. However, some residual noise can still be observed in the recovered image, indicated in the red rectangles. Compared with the other methods, the proposed NonLRTV method achieves the best visual performance. The restoration images of band 62 in the Pavia data set are presented in Fig. 4. This band was contaminated by the mixed noise of the Gaussian noise and stripes. It is shown that LRTV shows blurry image features. The LRMR, NAILRMA, WSN-LRMA, and the NonLRMA methods remove the noise and preserve the details to some degree, while the Gaussian noise is not removed completely. Among the methods under comparisons, the proposed NonLRTV performs the best in effectively removing the noise and preserving the image details. The quantitative comparisons between different methods are reported in Table II. We calculate the MPSNR and the MSSIM for the data set, as well as the PSNR and SSIM values for three different bands. The results show that the proposed NonLRTV method obtains the higher evaluation measurement values consistently, with over 10-dB improvement over the noisy image.

B. Real Data Experiments

In this section, in order to further investigate the utility of the proposed algorithm, we use three real-world HSI data sets in the experiments, including the airborne visible/infrared imaging spectrometer (AVIRIS) Indian Pines data set, the earth observing (EO-1) Hyperion data set, and the EO-1 Hyperion Australia data set.

1) AVIRIS Indian Pines Data Set: The Indian Pines data set was acquired by the NASA AVIRIS instrument over the Indian Pines test site in Northwestern Indiana in 1992. The size of the data set is $145 \times 145 \times 220$ [54]. The data set is mainly corrupted by water absorption, atmosphere, and so on.

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TABLE I

QUANTITATIVE EVALUATION OF DIFFERENT RESTORATION METHODS WITH HYDICE WASHINGTON DC MALL DATA SET

Noise Case	Evaluation index	Noisy	LRMR	LRTV	NAILRMA	WSN-LRMA	NonLRMA	NonLRTV
Overall	MPSNR	21.21	27.56	29.94	31.72	30.36	31.76	**33.28**
	MSSIM	0.3160	0.6638	0.7426	0.8388	0.7907	0.8403	**0.8949**
Band 2	PSNR	18.01	23.71	32.26	31.63	30.13	31.54	**34.65**
	SSIM	0.1644	0.4301	0.8025	0.8208	0.7627	0.8163	**0.9001**
Band 7	PSNR	22.12	30.89	29.66	32.04	31.49	32.98	**34.52**
	SSIM	0.3729	0.7925	0.7331	0.8553	0.8059	0.8557	**0.9085**
Band 15	PSNR	17.95	23.29	26.84	30.15	29.01	30.33	**31.46**
	SSIM	0.2953	0.5451	0.6712	0.8296	0.7844	0.8347	**0.8758**

TABLE II

QUANTITATIVE EVALUATION OF DIFFERENT RESTORATION METHODS WITH PAVIA CITY CENTER DATA SET

Noise Case	Evaluation index	Noisy	LRMR	LRTV	NAILRMA	WSN-LRMA	NonLRMA	NonLRTV
Overall	MPSNR	15.80	24.90	26.27	25.93	25.62	26.08	**27.58**
	MSSIM	0.2382	0.7387	0.8378	0.8136	0.7795	0.8277	**0.8455**
Band 10	PSNR	14.71	25.41	26.23	26.26	26.29	25.98	**26.67**
	SSIM	0.1440	0.7191	0.7980	0.8028	0.7972	0.7943	**0.8060**
Band 38	PSNR	13.12	19.71	26.26	24.16	24.28	24.42	**26.47**
	SSIM	0.1225	0.5096	0.8302	0.7991	0.8050	0.8085	**0.8446**
Band 62	PSNR	16.68	24.85	27.32	26.17	26.03	26.26	**27.57**
	SSIM	0.3353	0.7906	0.8702	0.8544	0.8548	0.8582	**0.8706**

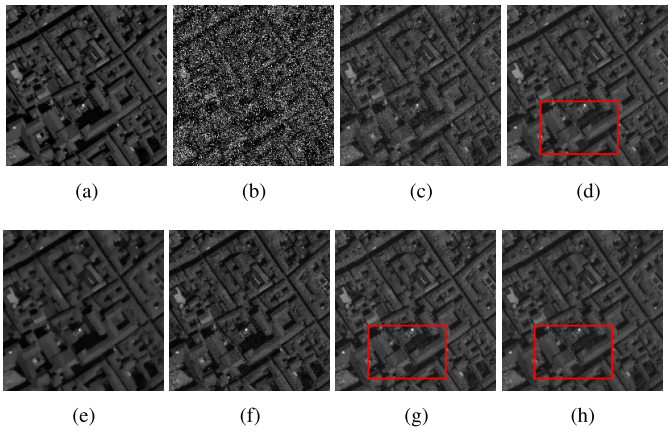


Fig. 3. Restoration results on simulated experiment: Pavia city center. (a) Original image of band 38. (b) Noisy image. Restored images obtained by (c) LRMR, (d) NAILRMA, (e) LRTV, (f) WSN-LRMA, (g) NonLRMA, and (h) proposed NonLRTV.

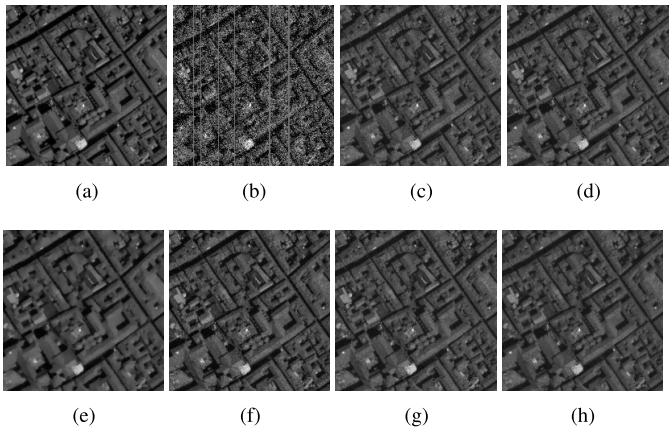


Fig. 4. Restoration results on simulated experiment: Pavia city center. (a) Original image of band 62. (b) Noisy image. Restored images obtained by (c) LRMR, (d) NAILRMA, (e) LRTV, (f) WSN-LRMA, (g) NonLRMA, and (h) proposed NonLRTV.

In order to evaluate the performance of the proposed NonLRTV method, we compare NonLRTV with LRMR, LRTV, NAILRMA, WSN-LRMA, and NonLRMA methods. Figs. 5 and 6 present the restored images for a moderate noise band 109 and a severe noise band 220, respectively. We observe that for band 109, the restored images by LRMR, NAILRMA, WSN-LRMA, and the NonLRMA methods contain obvious noise, with the image details lost, specifically in the area highlighted by the red rectangles. Both LRTV and NonLRTV methods are capable of removing most of the noise. However, LRTV cannot preserve edges and local detail information as well as NonLRTV, as indicated by red arrows. The results obviously reveal the effectiveness of the proposed NonLRTV method by using both nonconvex regularizations.

2) EO-1 Hyperion Australia Data Set: The EO-1 Hyperion L1R-level Australia data set was captured on December 4, 2010 with the location at Australia. The original image is of size $256 \times 3858 \times 242$ [55]. We remove the overlapping bands between visual near-infrared and shortwave infrared ranges, we use a subregion of size $400 \times 200 \times 150$ in the experiments. Figs. 7 and 8 present the restoration results of band 49 and band 124, respectively.

In Fig. 7, it is obvious to observe that the restored images by LRMR and NAILRMA methods contain obvious stripes noise. The LRTV method is able to handle the stripes noise well, whereas the image details are not well preserved, indicated by the red and the green arrows. WSN-LRMA and NonLRMA perform better in removing the noise, but the stripes remain in the recovered image. Compared with other methods, our proposed NonLRTV scheme has superior performance in removing most of the noise and preserving image edges and fine features well.

Fig. 8 presents the denoising results of a severe noise band, band 49. The original image and the restored images using different methods are shown in Fig. 8(a)–(g). In order to see the details more clearly, we show the magnified area

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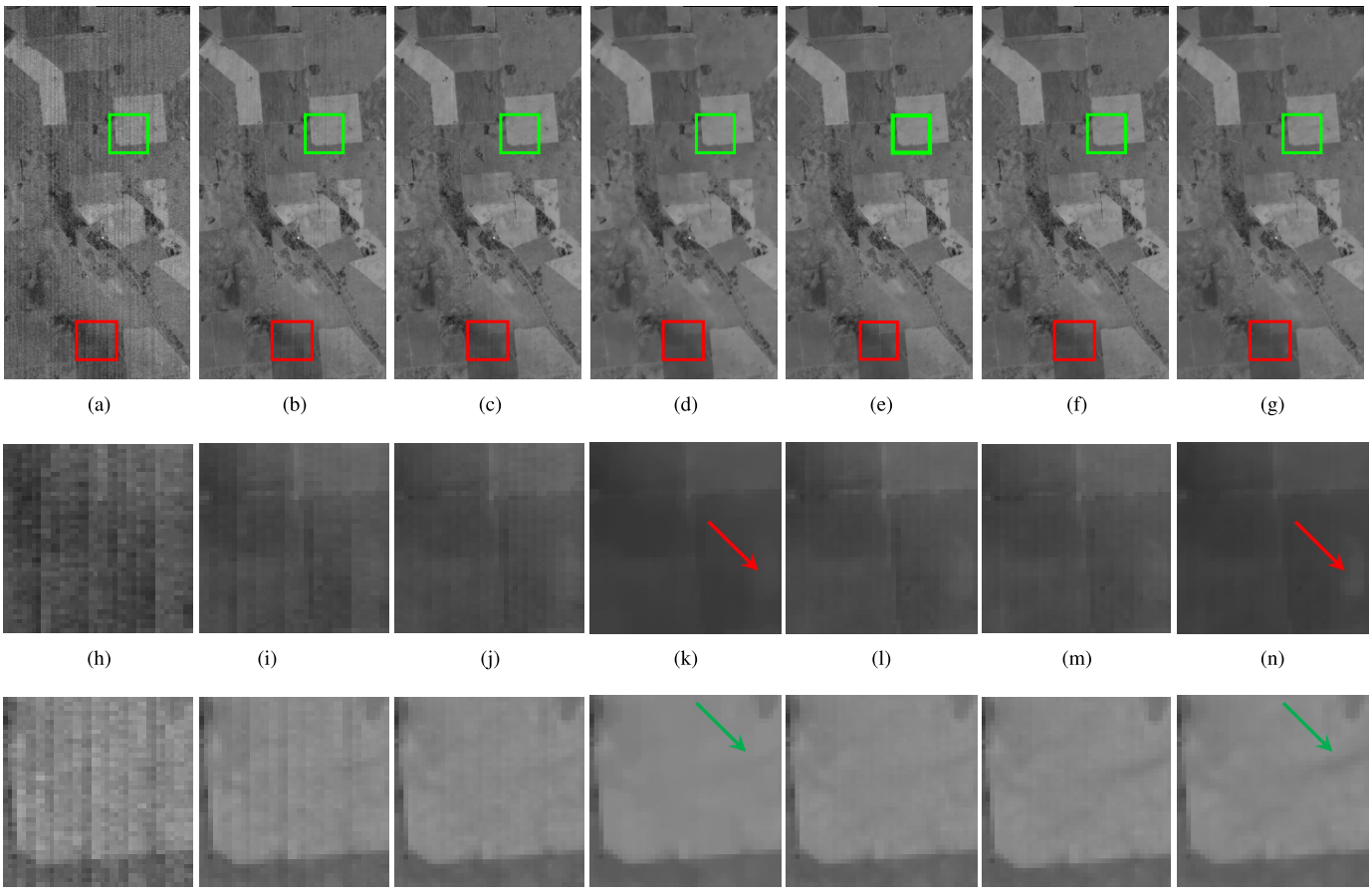
(a) (b) (c) (d) (e) (f) (g)

Fig. 5. Restoration results of band 109 in AVIRIS Indian Pines experiment. (a) Original image. Restored images obtained by (b) LRMR, (c) NAILRMA, (d) LRTV, (e) WSN-LRMA, (f) NonLRMA, and (g) proposed NonLRTV.



(a) (b) (c) (d) (e) (f) (g)

Fig. 6. Restoration results of band 220 in AVIRIS Indian Pines experiment. (a) Original image. Restored images obtained by (b) LRMR, (c) NAILRMA, (d) LRTV, (e) WSN-LRMA, (f) NonLRMA, and (g) proposed NonLRTV.



(a) (b) (c) (d) (e) (f) (g)

(h) (i) (j) (k) (l) (m) (n)

(o) (p) (q) (r) (s) (t) (u)

Fig. 7. Restoration results of band 124 in EO-1 Hyperion Australia data set. (a) Original image. (b)–(g) Restored images obtained by LRMR, NAILRMA, LRTV, WSN-LRMA, NonLRMA, and proposed NonLRTV, respectively. (h)–(n) Images of the magnified areas indicated by the red rectangles. (o)–(u) Images of the magnified areas indicated by the green rectangles.

in Fig. 8(h)–(n), indicated in the red rectangles. It is shown that the LRMR and NAILRMA methods are not able to recover the image effectively. LRTV remove the noise to some extent, but there is obvious Gaussian noise remained in the restored image, shown in Fig. 8(d) and (j). WSN-LRMA and NonLRMA show improved performance over the aforementioned three methods, while the stripes can still be noticed in the magnified restored image in Fig. 8(m), shown by the

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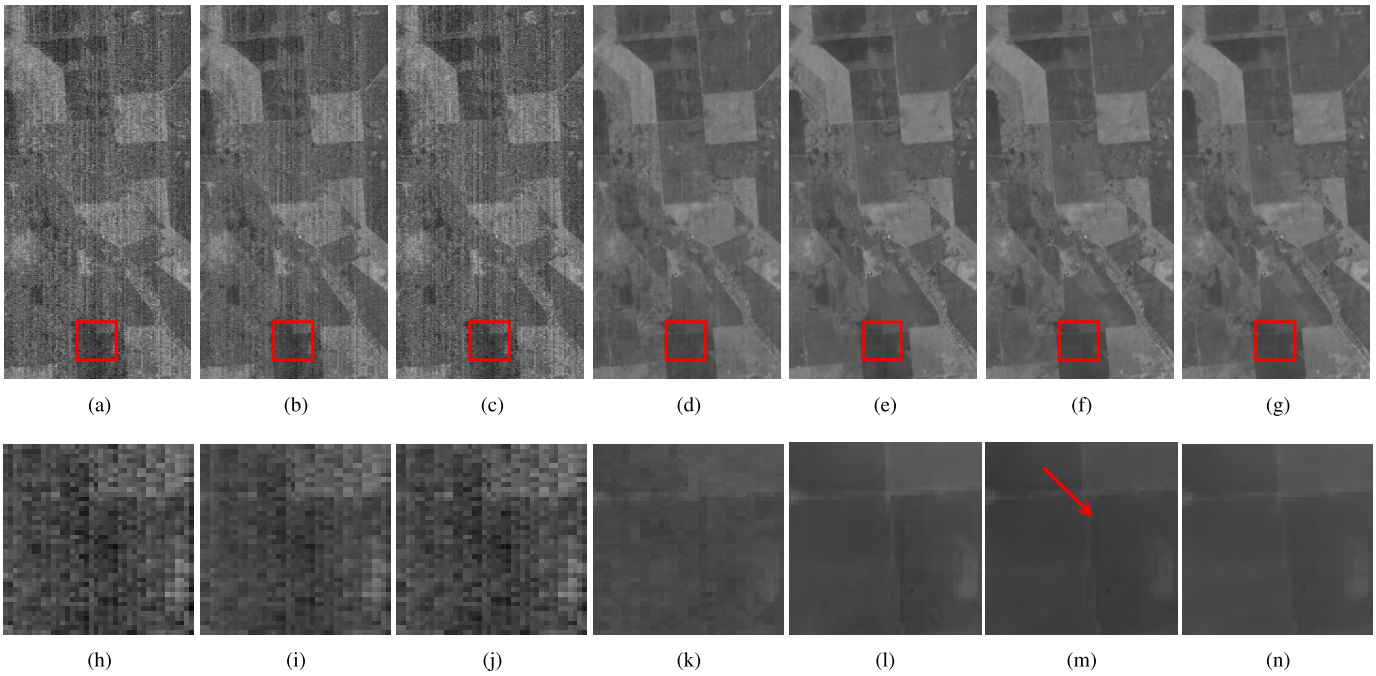


Fig. 8. Restoration results of band 49 in EO-1 Hyperion Australia data set. (a) Original image. (b)–(g) Restored images obtained by LRMR, NAILRMA, LRTV, WSN-LRMA, NonLRMA, and proposed NonLRTV, respectively. (h)–(n) Images of the magnified areas indicated by the red rectangles.

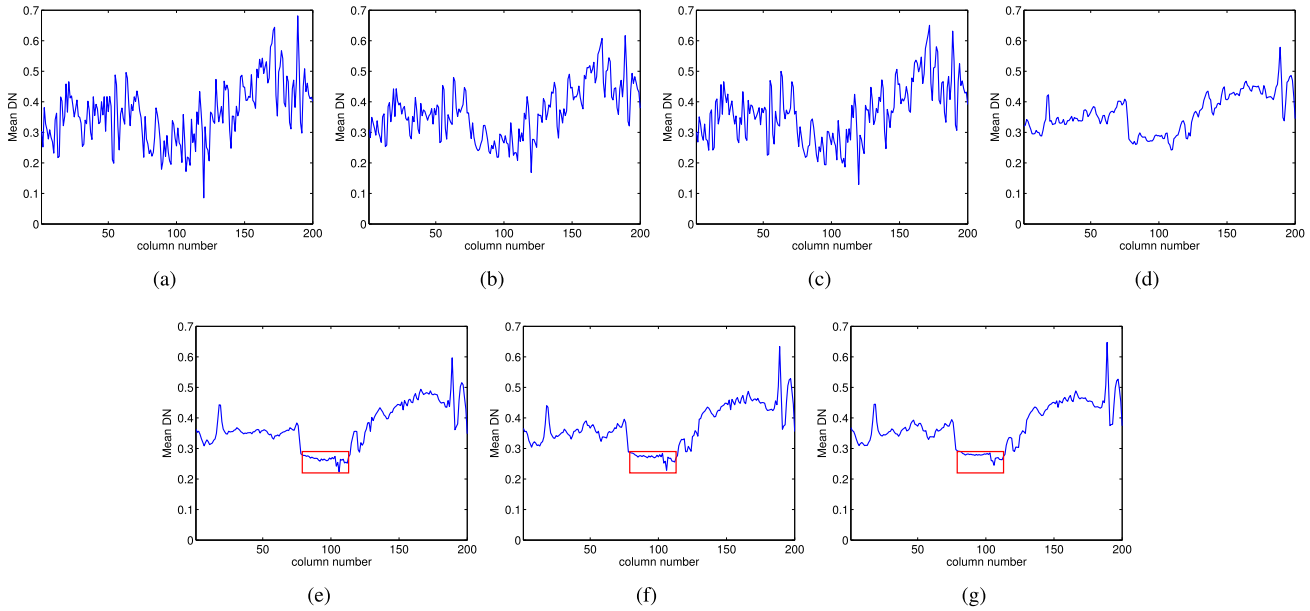


Fig. 9. Vertical mean profiles of band 49 in EO-1 Hyperion Australia data set. (a) Original band. (b) LRMR. (c) NAILRMA. (d) LRTV. (e) WSN-LRMA. (f) NonLRMA. (g) NonLRTV.

red arrow. We observe that our proposed NonLRTV provides the best results among the methods under comparisons. Due to the combined nonconvex spectral and spatial regularizations, the NonLRTV scheme can obtain more accurate restored HSI data. To better illustrate the utility of NonLRTV method, we display the vertical and horizontal mean profiles of band 49 in Figs. 9 and 10, respectively. We observe that the restored images by the LRMR, NAILRMA, and LRTV methods contain obvious noise. The rapid fluctuation in Fig. 10(f) indicates the remaining stripes and deadlines in the restored image by NonLRMA method. From all the recovered vertical and horizontal mean profiles, it is shown that the NonLRTV scheme provides the best restoration results in removing the mixed noise.

3) EO-1 Hyperion Data Set: In the third real data experiment, we use an EO-1 Hyperion data set. The size of the original image is $1000 \times 400 \times 242$ [56]. We remove the water absorption bands and use a subregion with the size of $200 \times 200 \times 166$. The Hyperion data set is mainly corrupted by Gaussian noise, stripes, and deadlines. In Figs. 11 and 12, we show the restoration results for band 1 and band 4, respectively. According to Fig. 11, we observe that the LRMR

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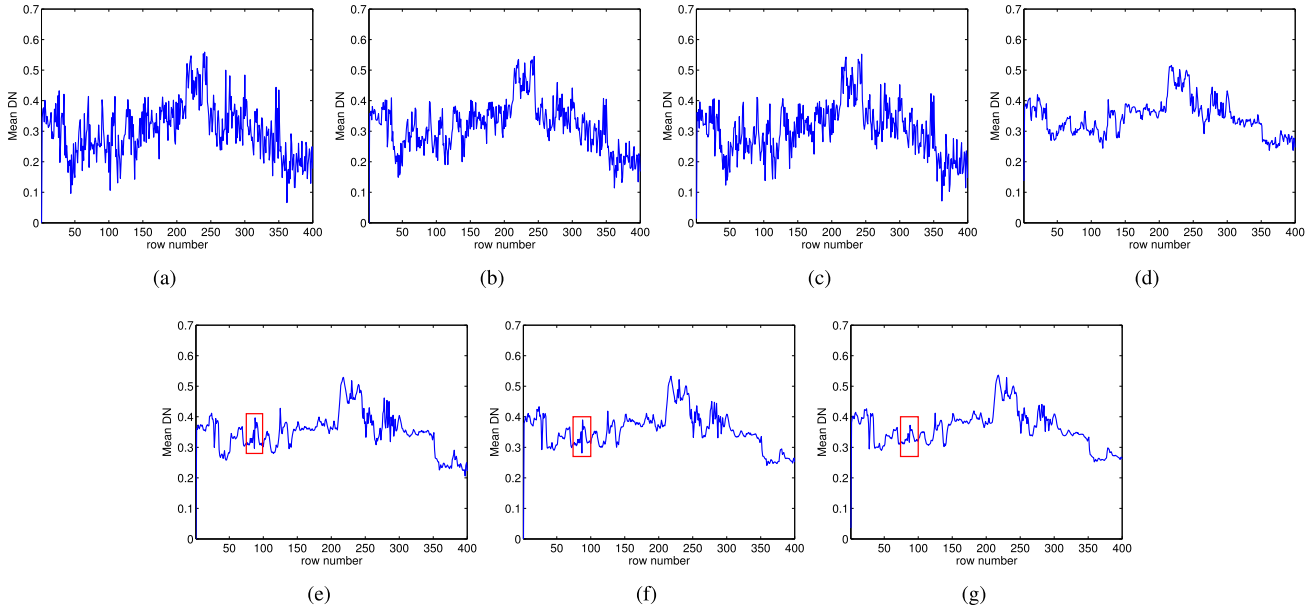


Fig. 10. Horizontal mean profiles of band 49 in EO-1 Hyperion Australia data set. (a) Original band. (b) LRMR. (c) NAILRMA. (d) LRTV. (e) WSN-LRMA. (f) NonLRMA. (g) NonLRTV.

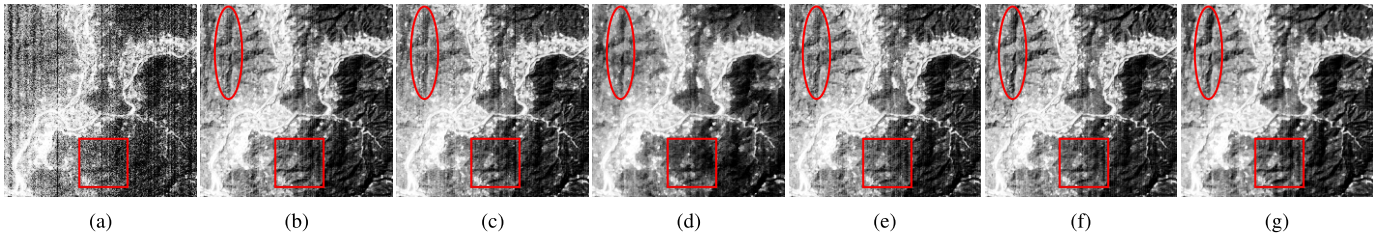


Fig. 11. Restoration results of band 1 in EO-1 Hyperion experiment. (a) Original image. Restored images obtained by (b) LRMR, (c) NAILRMA, (d) LRTV, (e) WSN-LRMA, (f) NonLRMA, and (g) proposed NonLRTV.

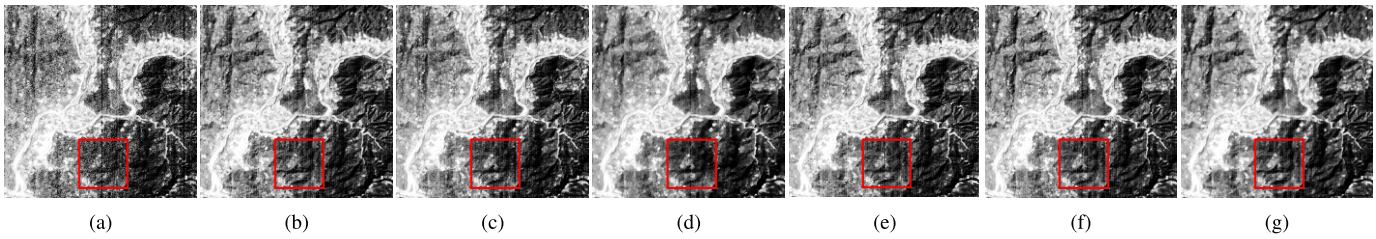


Fig. 12. Restoration results of band 4 in EO-1 Hyperion experiment. (a) Original image. Restored images obtained by (b) LRMR, (c) NAILRMA, (d) LRTV, (e) WSN-LRMA, (f) NonLRMA, and (g) proposed NonLRTV.

and NAILRMA methods are able to suppress the Gaussian noise. However, the stripes and deadlines are obviously presented in the restored images, shown in Fig. 11(b) and (c). LRTV scheme removes both of the Gaussian noise and the stripes without effectively preserving the image details and edges, which can be observed in the highlighted rectangle and the ellipse area. The NonLRMA method performs better in preserving the image features. However, compared with the proposed NonLRTV method, the sparsity in the spatial domain is not fully exploited, resulting in noticeable stripes and deadlines noise in the recovered image. Similarly, we observe that the proposed NonLRTV scheme outperforms the other methods under comparisons in Fig. 12 in effectively removing the mixed noise while preserving the image details. It is seen that, in Fig. 12(b), (c), (e), and (f), the LRMR, NAILRMA, WSN-LRMA, and NonLRMA methods fail to restore some stripes. The reason is that these four methods mainly focus on the spectral correlation of the HSI data set, which leads to suboptimal performance compared to the NonLRTV scheme. LRTV exploits both the spectral correlation and spatial sparsity. However, the convex penalties are not able to capture the image features, which result in blurred image details, indicated in the red rectangle in Fig. 12(d).

The vertical and the horizontal mean profiles of band 1 are plotted in Figs. 13 and 14, respectively. We observe that the original image contains stripes and deadlines, which are indicated by the rapid fluctuations in the mean profiles, shown in Figs. 13(a) and 14(a). It is shown that the LRMR, NAILRMA, WSN-LRMA, and NonLRMA schemes are not able to remove the stripes effectively, which are indicated in

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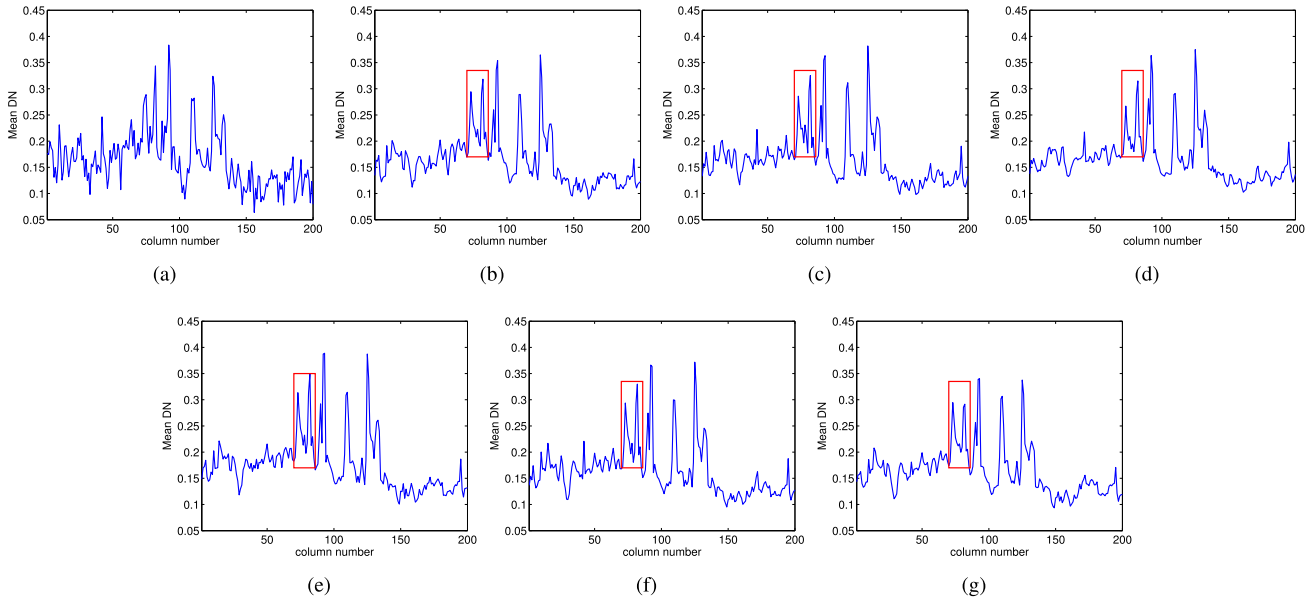


Fig. 13. Vertical mean profiles of band 1 in EO-1 Hyperion data set. (a) Original band. (b) LRMR. (c) NAILRMA. (d) LRTV. (e) WSN-LRMA. (f) NonLRMA. (g) NonLRTV.

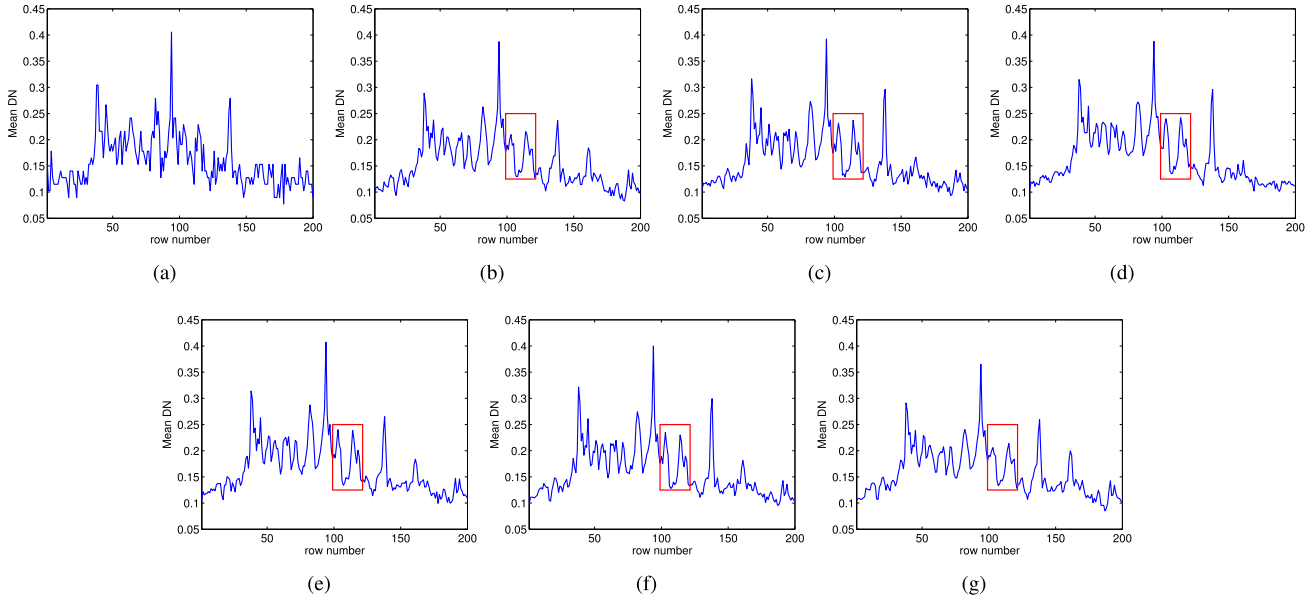


Fig. 14. Horizontal mean profiles of band 1 in EO-1 Hyperion data set. (a) Original band. (b) LRMR. (c) NAILRMA. (d) LRTV. (e) WSN-LRMA. (f) NonLRMA. (g) NonLRTV.

the red rectangles in Fig. 13(b), (c), (e), and (f), where the rapid fluctuations are not obviously suppressed. The LRTV method displays oversmooth mean profiles, which explains the blurred image details in the restored image. The proposed NonLRTV method can effectively remove mixed noise, compared with the other methods.

C. Model Analysis

In this section, we discuss the parameter settings and analyze the sensitivity of the parameters in the proposed model. In addition, we analyze the computational complexity and the convergence of the algorithm. We also report the running time of different HSI denoising methods. The experiments used for parameter settings are based on the HSI data set of Pavia city center in Section IV-A2.

1) Sensitivity Analysis of Parameters p_1 and p_2: To determine the optimal p_1 and p_2 parameters for the nonconvex low-rank and nonconvex sparsity penalties, we plot the MPSNR values of the recovered HSI data set as function of p_1 and p_2 values shown in Fig. 15. We choose p_1 and p_2 from 0.1 to 0.9. The MPSNR values using different combinations of p_1 and p_2 values are reported in Table III. We observe that when $p_1, p_2 \in (0.5, 0.9]$, as p_1 and p_2 decrease, the results get improved significantly. However, when $p_1, p_2 \in (0, 0.5]$, the results show no obvious difference. Thus, we choose $p_1 = p_2 = 0.5$ in all the experiments.

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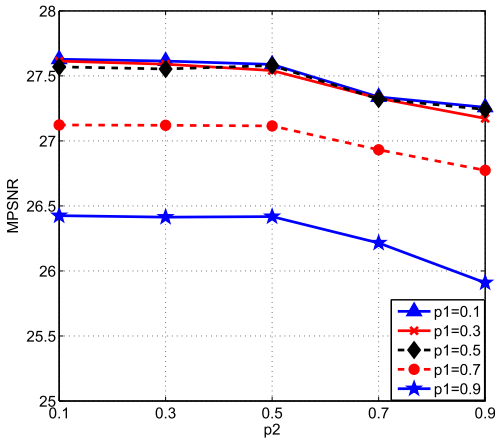


Fig. 15. Sensitivity analysis of parameters p_1 and p_2. The plot shows the change of MPSNR values of the recovered HSI data set using different combinations of p_1 and p_2 parameters.

TABLE III

MPSNR VALUES OF THE NONLRTV RESULTS USING DIFFERENT p_1 AND p_2 PARAMETERS

p_1 \ p_2	**0.1**	**0.3**	**0.5**	**0.7**	**0.9**
0.1	27.63	27.61	27.59	27.34	27.26
0.3	27.61	27.59	27.54	27.33	27.17
0.5	27.57	27.55	27.58	27.32	27.24
0.7	27.12	27.12	27.11	26.93	26.77
0.9	26.42	26.41	26.42	26.21	25.91

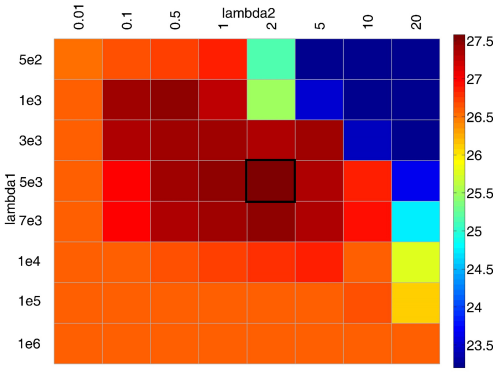
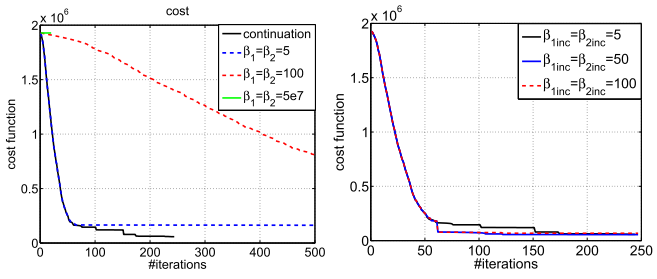


Fig. 16. MPSNR values of the recovered HSI data set as function of the parameters λ_1 and λ_2. The region in bold indicates the maximum MPSNR value, where the optimal parameters are chosen.

2) Sensitivity Analysis of Parameters λ_1 and λ_2: The regularization parameters λ_1 and λ_2 balance the nonconvex low-rank penalty and the nonconvex sparsity penalty. In order to investigate how the MPSNR values of the restored images behave as function of λ_1 and λ_2, we plot the parameters optimization results in Fig. 16. The region in bold corresponds to the maximum MPSNR value, where the optimal parameters were chosen, i.e., $\lambda_1 = 5e3$, $\lambda_2 = 2$. The plot indicates that the MPSNR value has a global maximum for different values of λ_1 and λ_2. We observe that for both simulated and real data sets, when $\lambda_1 \in [3e3, 1e4]$ and $\lambda_2 \in [2, 8]$, the proposed method obtains the optimal performance.

3) Convergence of the Algorithm: In the optimization algorithm, we use the continuation scheme to guarantee the convergence of the algorithm. We now study the effectiveness of the parameters β_1, β_2, β_{inc1}, and β_{inc2} on the convergence performance. In Fig. 17(a), we plot the evolution of the cost function with respect to the number of iterations using different values of β_1 and β_2. We observe that since smaller values of β_1 and β_2 make (16) poor approximation to the original optimization problem (9), they result in faster convergence but higher cost. On the other hand, larger values of β_1 and β_2 make better approximation to (9), but yield slow convergence and higher cost. However, using the continuation scheme, where we initialize $\beta_1 = \beta_2 = 5$ and gradually increase them by $\beta_{inc1} = \beta_{inc2} = 5$, obtains the optimal result and fastest convergence speed. We illustrate the effectiveness of β_{inc1} and β_{inc2} on the convergence shown in Fig. 17(b). It is observed that different choices of β_{inc1} and β_{inc2} convergence to similar results, indicating that the algorithm is not very sensitive to the values of β_{inc1} and β_{inc2}.



(a) (b)

Fig. 17. Sensitivity analysis of the parameters β_1, β_2, β_{inc1}, and β_{inc2}. (a) Cost function to iterations using different β_1 and β_2. (b) Cost function to iterations using different β_{inc1} and β_{inc2}.

TABLE IV

MPSNR AND MSSIM VALUES OF THE NONLRTV RESULTS USING DIFFERENT PATCH SIZES m (WITH THE STEP SIZE s BEING 4)

Patch Size m	**10**	**12**	**16**	**20**
MPSNR	27.08	27.14	27.31	**27.58**
MSSIM	0.8226	0.8264	0.8384	**0.8455**
Patch Size m	**24**	**30**	**40**	**50**
MPSNR	27.25	27.13	27.11	27.08
MSSIM	0.8295	0.8249	0.8238	0.8218

TABLE V

MPSNR AND MSSIM VALUES OF THE NONLRTV RESULTS USING DIFFERENT STEP SIZES s (WITH THE PATCH SIZE m BEING 20)

Step Size s	**2**	**4**	**8**	**16**	**20**	**24**
MPSNR	**27.69**	27.58	27.50	27.39	26.15	23.22
MSSIM	**0.8489**	0.8455	0.8403	0.8368	0.8135	0.7628

4) Analysis of Spatial Patch Size m and Step Size s: In this section, given the optimal values of parameters p_1, p_2, λ_1, λ_2, β_1, and β_2, we investigate the influence of the spatial patch size m and the step size s on the recovered image quality. In Table IV, we plot the MPSNR values and the MSSIM values of the recovered images obtained by the proposed NonLRTV with different patch sizes with the step size being 4. Table V shows the MPSNR and MSSIM values of the restored images using different step sizes with the patch size being 20. Based on the results, we observe that the proposed NonLRTV is not very sensitive to the patch size. As for the step size, when s is less than m, we obtain similar results using different

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TABLE VI
COMPARISONS OF RUNNING TIME OF DIFFERENT METHODS (IN SECONDS)

Method	LRMR	NAILRMA	LRTV
Running Time	180	52	98
Method	WSN-LRMA	NonLRMA	NonLRTV
Running Time	3695	113	286

step sizes. However, when s is larger than the patch size, the method presents decreased performance. Considering the computational burden and the image recovery performance, we set $m = 20$ and $s = 4$ in simulated data experiments, and $m = 50$ and $s = 16$ in real data experiments.

5) Analysis of Computational Complexity: In this section, we analyze the computational complexity of the proposed algorithm. For each patch in an HSI data with the patch size $m \times m \times K$ and the step size s, in each iteration of Algorithm 1, the cost of the NonLRTV method lies in the subproblems of updating $\mathbf{H}$, $\mathbf{P}$, and $\mathbf{T}_i$. The computational demand for subproblem (17) is $\mathcal{O}(m^2K)$. For subproblem (19), the SVD and the shrinkage step dominate the main computation. Thus, the complexity of updating $\mathbf{P}$ is $\mathcal{O}(\min\{m^4K, m^2K^2\} + m^2K)$. The cost of updating $\mathbf{T}_i$ is $\mathcal{O}(3m^2K)$. The number of patches is in inverse proportion to s^2. Thus, the complexity of the proposed NonLRTV method for the HSI data is $\mathcal{O}(\min\{m^4K, m^2K^2\}/s^2 + 5m^2K/s^2)$.

6) Running Time: The running time using different methods are compared in Table VI. We observe that the NAILRMA method is the fastest approach among all the compared methods. However, NAILRMA method shows limited restoration performance. Among the three nonconvex models under comparisons, which are able to obtain more satisfactory results, the running time of the proposed NonLRTV method is relatively higher than NonLRMA but significantly lower than WSN-LRMA. Although the proposed method requires more time than NonLRMA, the previous experiments demonstrate that it is able to provide significant improvement in image quality.

V. CONCLUSION

In this article, we propose a NonLRTV method by combing the nonconvex low-rank penalty and the nonconvex TV for the restoration of HSI data sets with mixed noise. By using the nonconvex spectral and spatial penalties, we are capable of obtaining a more accurate approximation of the original image, with intrinsic spatial–spectral correlations. In order to effectively solve the nonconvex regularized optimization problem, we introduce a fast iterative variable splitting-based algorithm. The experiments demonstrate the improvement in image quality provided by the proposed algorithm.

REFERENCES

[1] A. Plaza *et al.*, "Recent advances in techniques for hyperspectral image processing," *Remote Sens. Environ.*, vol. 113, pp. S110–S122, Sep. 2009.

[2] D. Haboudane, J. R. Miller, E. Pattey, P. J. Zarco-Tejada, and I. B. Strachan, "Hyperspectral vegetation indices and novel algorithms for predicting green LAI of crop canopies: Modeling and validation in the context of precision agriculture," *Remote Sens. Environ.*, vol. 90, no. 3, pp. 337–352, 2004.

[3] H. Zhang, W. He, L. Zhang, H. Shen, and Q. Yuan, "Hyperspectral image restoration using low-rank matrix recovery," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 8, pp. 4729–4743, Aug. 2014.

[4] M. Elad and M. Aharon, "Image denoising via sparse and redundant representations over learned dictionaries," *IEEE Trans. Image Process.*, vol. 15, no. 12, pp. 3736–3745, Dec. 2006.

[5] K. Dabov, A. Foi, V. Katkovnik, and K. Egiazarian, "Image denoising by sparse 3-D transform-domain collaborative filtering," *IEEE Trans. Image Process.*, vol. 16, no. 8, pp. 2080–2095, Aug. 2007.

[6] J. Mairal, F. Bach, J. Ponce, G. Sapiro, and A. Zisserman, "Non-local sparse models for image restoration," in *Proc. IEEE 12th Int. Conf. Comput. Vis.*, Sep. 2009, pp. 2272–2279.

[7] G. Chen and S.-E. Qian, "Denoising of hyperspectral imagery using principal component analysis and wavelet shrinkage," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 3, pp. 973–980, Mar. 2011.

[8] Q. Yuan, L. Zhang, and H. Shen, "Hyperspectral image denoising with a spatial–spectral view fusion strategy," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 5, pp. 2314–2325, May 2014.

[9] H. Othman and S.-E. Qian, "Noise reduction of hyperspectral imagery using hybrid spatial-spectral derivative-domain wavelet shrinkage," *IEEE Trans. Geosci. Remote Sens.*, vol. 44, no. 2, pp. 397–408, Feb. 2006.

[10] Y. Hu, S. G. Lingala, and M. Jacob, "A fast majorize–minimize algorithm for the recovery of sparse and low-rank matrices," *IEEE Trans. Image Process.*, vol. 21, no. 2, pp. 742–753, Aug. 2012.

[11] E. J. Candès, X. Li, Y. Ma, and J. Wright, "Robust principal component analysis?" *J. ACM*, vol. 58, no. 3, p. 11, May 2011.

[12] C.-I. Chang and Q. Du, "Interference and noise-adjusted principal components analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 37, no. 5, pp. 2387–2396, Sep. 1999.

[13] J. Wright, A. Ganesh, S. Rao, Y. Peng, and Y. Ma, "Robust principal component analysis: Exact recovery of corrupted low-rank matrices via convex optimization," in *Proc. Adv. Neural Inf. Process. Syst.*, 2009, pp. 2080–2088.

[14] W. He, H. Zhang, L. Zhang, and H. Shen, "Hyperspectral image denoising via noise-adjusted iterative low-rank matrix approximation," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 6, pp. 3050–3061, Jun. 2015.

[15] Q. Xie *et al.*, "Multispectral images denoising by intrinsic tensor sparsity regularization," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2016, pp. 1692–1700.

[16] H. Fan, Y. Chen, Y. Guo, H. Zhang, and G. Kuang, "Hyperspectral image restoration using low-rank tensor recovery," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 10, no. 10, pp. 4589–4604, Oct. 2017.

[17] F. Xu, Y. Chen, C. Peng, Y. Wang, X. Liu, and G. He, "Denoising of hyperspectral image using low-rank matrix factorization," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 7, pp. 1141–1145, Jul. 2017.

[18] J. Xue, Y. Zhao, W. Liao, and S. G. Kong, "Joint spatial and spectral low-rank regularization for hyperspectral image denoising," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 4, pp. 1940–1958, Apr. 2018.

[19] L. I. Rudin, S. Osher, and E. Fatemi, "Nonlinear total variation based noise removal algorithms," *Phys. D, Nonlinear Phenomena*, vol. 60, nos. 1–4, pp. 259–268, 1992.

[20] T. Chan, S. Esedoglu, F. Park, and A. Yip, "Recent developments in total variation image restoration," *Math. Models Comput. Vis.*, vol. 17, no. 2, pp. 1–18, 2005.

[21] W. K. Allard, "Total variation regularization for image denoising, II. Examples," *SIAM J. Imag. Sci.*, vol. 1, no. 4, pp. 400–417, 2008.

[22] Q. Yuan, L. Zhang, and H. Shen, "Hyperspectral image denoising employing a spectral–spatial adaptive total variation model," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 10, pp. 3660–3677, Oct. 2012.

[23] H. K. Aggarwal and A. Majumdar, "Hyperspectral image denoising using spatio-spectral total variation," *IEEE Geosci. Remote Sens. Lett.*, vol. 13, no. 3, pp. 442–446, Mar. 2016.

[24] Y. Chang, L. Yan, H. Fang, and H. Liu, "Simultaneous destriping and denoising for remote sensing images with unidirectional total variation and sparse representation," *IEEE Geosci. Remote Sens. Lett.*, vol. 11, no. 6, pp. 1051–1055, Jun. 2014.

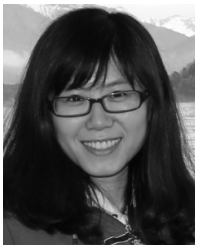
[25] H. Zhang, "Hyperspectral image denoising with cubic total variation model," *ISPRS Ann. Photogram., Remote Sens. Spatial Inf. Sci.*, vol. 7, pp. 95–98, Jul. 2012.

[26] W. He, H. Zhang, L. Zhang, and H. Shen, "Total-variation-regularized low-rank matrix factorization for hyperspectral image restoration," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 1, pp. 178–188, Jan. 2016.

[27] S. Gu, L. Zhang, W. Zuo, and X. Feng, "Weighted nuclear norm minimization with application to image denoising," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2014, pp. 2862–2869.

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[28] Y. G. Peng, J. L. Suo, Q. H. Dai, and W. L. Xu, "Reweighted low-rank matrix recovery and its application in image restoration," *IEEE Trans. Cybern.*, vol. 44, no. 12, pp. 2418–2430, Dec. 2014.
[29] Z. Wu, Q. Wang, Z. Wu, and Y. Shen, "Total variation-regularized weighted nuclear norm minimization for hyperspectral image mixed denoising," *J. Electron. Imag.*, vol. 25, no. 1, 2016, Art. no. 013037.
[30] Y. Chen, S. Wang, and Y. Zhou, "Tensor nuclear norm-based low-rank approximation with total variation regularization," *IEEE J. Sel. Topics Signal Process.*, vol. 12, no. 6, pp. 1364–1377, Oct. 2018.
[31] Y. Wang, J. Peng, Q. Zhao, D. Meng, Y. Leung, and X.-L. Zhao, "Hyperspectral image restoration via total variation regularized low-rank tensor decomposition," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 4, pp. 1227–1243, Apr. 2018.
[32] L. Sun, Y. Zheng, and B. Jeon, "Hyperspectral restoration employing low rank and 3D total variation regularization," in *Proc. Int. Conf. Progr. Inform. Comput. (PIC)*, Dec. 2016, pp. 326–329.
[33] H. Zou, "The adaptive lasso and its oracle properties," *J. Amer. Statist. Assoc.*, vol. 101, no. 476, pp. 1418–1429, Dec. 2006.
[34] R. Chartrand and V. Staneva, "Restricted isometry properties and nonconvex compressive sensing," *Inverse Problems*, vol. 24, no. 3, Jun. 2008, Art. no. 035020.
[35] Z. Xu, H. Zhang, Y. Wang, X. Chang, and Y. Liang, "$L_{1/2}$ regularization," *Sci. China Inf. Sci.*, vol. 53, no. 6, pp. 1159–1169, 2010.
[36] C. Lu, J. Tang, S. Yan, and Z. Lin, "Nonconvex nonsmooth low rank minimization via iteratively reweighted nuclear norm," *IEEE Trans. Image Process.*, vol. 25, no. 2, pp. 829–839, Feb. 2016.
[37] S. G. Lingala, Y. Hu, E. DiBella, and M. Jacob, "Accelerated dynamic MRI exploiting sparsity and low-rank structure: Kt SLR," *IEEE Trans. Med. Imag.*, vol. 30, no. 5, pp. 1042–1054, May 2011.
[38] Y. Xie, Y. Qu, D. Tao, W. Wu, Q. Yuan, and W. Zhang, "Hyperspectral image restoration via iteratively regularized weighted schatten p-norm minimization," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 8, pp. 4642–4659, Apr. 2016.
[39] Y. Chen, Y. Guo, Y. Wang, D. Wang, C. Peng, and G. He, "Denoising of hyperspectral images using nonconvex low rank matrix approximation," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 9, pp. 5366–5380, Sep. 2017.
[40] D. Geman and C. Yang, "Nonlinear image recovery with half-quadratic regularization," *IEEE Trans. Signal Process.*, vol. 4, no. 7, pp. 932–946, Jul. 1995.
[41] J. Yang, Y. Zhang, and W. Yin, "A fast TVL1-L2 minimization algorithm for signal reconstruction from partial Fourier data," CAAM, Rice Univ., Houston, TX, USA, Tech. Rep. TR08-27, 2008.
[42] Y. Wang, J. Yang, W. Yin, and Y. Zhang, "A new alternating minimization algorithm for total variation image reconstruction," *SIAM J. Imag. Sci.*, vol. 1, no. 3, pp. 248–272, Aug. 2008.
[43] N. Meinshausen and B. Yu, "Lasso-type recovery of sparse representations for high-dimensional data," *Ann. Statist.*, vol. 37, no. 1, pp. 246–270, 2009.
[44] Z.-B. Xu, H.-L. Guo, Y. Wang, and H. Zhang, "Representative of $L_{1/2}$ regularization among L_q $(0< q \leq 1)$ regularizations: An experimental study based on phase diagram," *Acta Automat. Sinica*, vol. 38, no. 7, pp. 1225–1228, 2012.
[45] R. Chartrand, "Exact reconstruction of sparse signals via nonconvex minimization," *IEEE Signal Process. Lett.*, vol. 14, no. 10, pp. 707–710, Oct. 2007.
[46] C. Lu, J. Tang, S. Yan, and Z. Lin, "Generalized nonconvex nonsmooth low-rank minimization," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2014, pp. 4130–4137.
[47] J. Yang and Y. Zhang, "Alternating direction algorithms for ℓ_1-problems in compressive sensing," *SIAM J. Sci. Comput.*, vol. 33, no. 1, pp. 250–278, 2011.
[48] J.-F. Cai, E. J. Candès, and Z. Shen, "A singular value thresholding algorithm for matrix completion," *SIAM J. Optim.*, vol. 20, no. 4, pp. 1956–1982, 2010.
[49] W. Zuo, D. Meng, L. Zhang, X. Feng, and D. Zhang, "A generalized iterated shrinkage algorithm for non-convex sparse coding," in *Proc. IEEE Int. Conf. Comput. Vis.*, Jun. 2013, pp. 217–224.
[50] R. Chartrand, "Fast algorithms for nonconvex compressive sensing: MRI reconstruction from very few data," in *Proc. IEEE Int. Symp. Biomed. Imag., Nano Macro*, Jun. 2009, pp. 262–265.
[51] J. Nocedal and S. Wright, *Numerical Optimization*, 2nd ed. New York, NY, USA: Springer-Verlag, 2006.
[52] *HYDICE Image of Washington DC Mall*. Accessed: Aug. 10, 2018. [Online]. Available: https://engineering.purdue.edu/biehl/MultiSpec/hyperspectral.html
[53] *ROSIS-03 Image of Pavia City Center*. Accessed: Aug. 10, 2018. [Online]. Available: http://www.opairs.aero/rosis_en.html
[54] *AVIRIS Indian Pines Image*. Accessed: Jul. 15, 2018. [Online]. Available: https://engineering.purdue.edu/~biehl/MultiSpec/hyperspectral.html
[55] *EO-1 Hyperion Australia Images*. Accessed: Jul. 15, 2018. [Online]. Available: http://remote-sensing.nci.org.au/
[56] *The EO-1 Hyperion Imaging*. Accessed: Jul. 15, 2018. [Online]. Available: http://www.gscloud.cn/



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